

5 (a)	work done in moving <u>unit</u> (positive) charge	M1	[2]
	from infinity to the point and charge is positive	A1	
(b) (i)	$V = Q / 4\pi\epsilon_0 r$ where ϵ_0 is permittivity (of free space)	B1	[3]
(ii)	$C = Q / V$	B1	
(iii)	<u>hence</u> $C = 4\pi\epsilon_0 r$	B1	
(c) (i) 1	$C = 4\pi \times 8.85 \times 10^{-11} \times 0.15$ $= 1.67 \times 10^{-5} \mu\text{F}$	B1	[4]
2	energy = $\frac{1}{2}CV^2$ or $\frac{1}{2}QV$	C1	
	potential = $(2.0 \times 10^{-6}) / (1.67 \times 10^{-11}) = 1.2 \times 10^5 \text{ V}$	C1	
	energy = $\frac{1}{2} \times 1.67 \times 10^{-11} \times (1.2 \times 10^5)^2$ $= 0.12 \text{ J}$	A1	